

Functions

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The notion of correspondence is encountered frequently in every day life.

To say that one quantity is a function of another means that the first quantity depends on the second in some specific way e.g.

$$\text{Circumference of a circle} = 2\pi r.$$

Definition: Let X and Y be two sets.

By a function f from X to Y , we mean a rule or relation which assigns to each x in X (i.e. $x \in X$) a unique element $f(x)$ in Y (i.e. $f(x) \in Y$).

$$\text{i.e. } f: X \xrightarrow{\text{mapping}} Y$$

The set X is called the domain of f , and is denoted by $\text{Dom}(f)$.

The set $y = f(x)$ is called the range of f , denoted by $\text{Ran}(f)$.

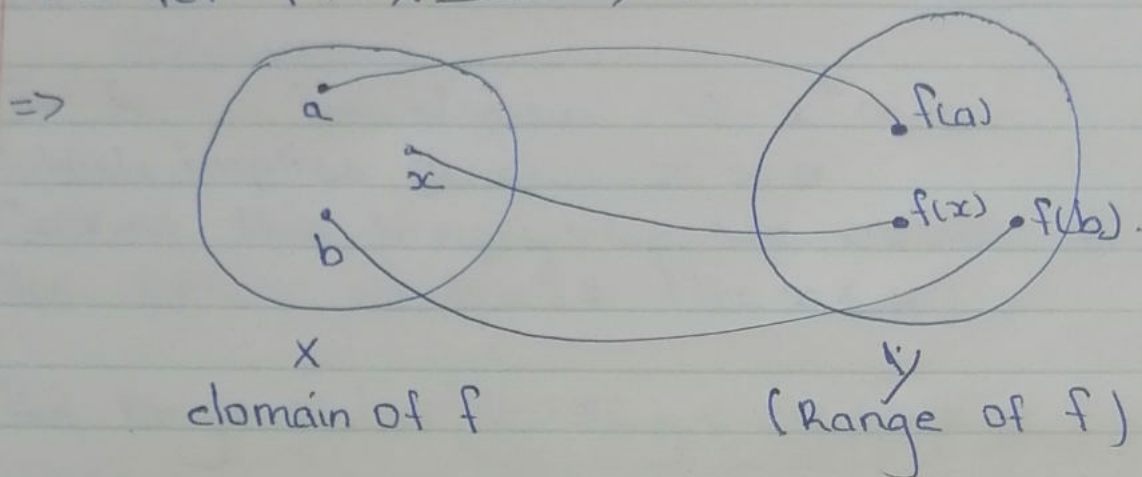
This range of f is also called the image of x under f .

By writing a formula $y = f(x)$, we call x the independent variable and y dependent variable.

The set of all values for which $f(x)$ is defined is the domain of f and the values $y = f(x)$ where $x \in \text{Dom}(f)$ is the range of f .

The graph of the function f is the set of all points in the plane of the form $(x, f(x))$, where $x \in \text{Dom}(f)$.

i.e. for $f: X \longrightarrow Y$



NB: The Largest domain of any given function is $\mathbb{R}$ (i.e. $-\infty, \infty$).

Example:

State the domain and range of each of the following functions

(i) $f(x) = 3x - 2$.

Solution

The function $f(x) = 3x - 2$ is defined for any Real number $\Rightarrow \text{Dom}(f) = \mathbb{R}$

Again the range of is the Set of Real numbers.

(ii) $g(x) = 2x^2 + 1$

Solution

The function $g(x)$ is defined for all values of real numbers i.e. $\text{Dom}(f) = \mathbb{R}$

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The range of $g(x)$ is the set of real numbers that are greater or equal to 1.
 i.e. $\text{Ran}(g) = \{y: y \geq 1, y \in \mathbb{R}\}$.

(iii) $h(x) = \sqrt{16 - x^2}$

Solution

For h to be defined, $16 - x^2 \geq 0$

which implies $(4 - x)(4 + x) \geq 0$.

Solving this inequality, we obtain that the domain of $h = \{x: -4 \leq x \leq 4\}$.

The range of $h = \{y: 0 \leq y \leq 4, y \in \mathbb{R}\}$.

(iv) $k(x) = \frac{x}{1 - x}$

Solution

Notice at $x = 1$, $k(1) = \frac{1}{0}$ which is undefined

$\Rightarrow k(x)$ is defined for all values of $x \in \mathbb{R}$ except at $x = 1$

i.e. $\text{Dom}(f) = \{x: x \neq 1, x \in \mathbb{R}\}$.

Range of k is the $\{y: y \neq 1, y \in \mathbb{R}\}$.

(v) $f(x) = x^2$

Solution

The function $f(x)$ is defined for all values of x in $\mathbb{R}$ i.e. domain $f = \{\mathbb{R}\}$.

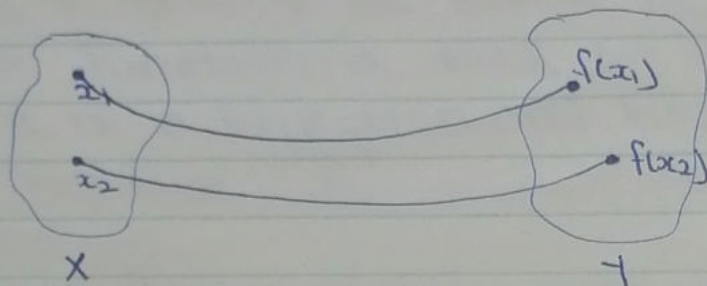
The range of f is the set of whole numbers ~~or~~ the set of positive integers.
 i.e. $\text{Ran}(f) = \{y: y \geq 0, y \in \mathbb{Z}\}$.

Definition A.

A function $f: X \rightarrow Y$ is one-to-one or injective if, for every pair $x_1, x_2 \in X$, $x_1 \neq x_2$, implies that $f(x_1) \neq f(x_2)$ (or equivalently, if $f(x_1) = f(x_2)$, then $x_1 = x_2$).

i.e.

$f: X \xrightarrow{\text{mapping}} Y$



e.g. $f(x) = 3x + 2$ is one-to-one (i.e. 1-1).
 $f(x) = x^2 + 5$ is one-to-one.

Definition B.

A function $f: X \rightarrow Y$ is onto or Surjective if, $y \in Y$ is of the form $f(x)$ for some $x \in X$. i.e. f is onto if $\text{Ran}(f) = \text{Whole of } Y$.

e.g. $f(x) = 2x$ is an onto function
 $f(x) = x^2$ is ~~not~~ onto function

Definition C:

A function $f: X \rightarrow Y$ is a bijection if it is 1-1 and onto

e.g. $f(x) = 2x$ is both onto and 1-1.
 and hence a bijection.

Note: for all the function under consideration the assumption is that we have the mapping $f: \mathbb{R} \rightarrow \mathbb{R}$

Definition D:

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If a function is a bijection, then there exists an inverse function

$$f^{-1}: Y \rightarrow X$$

$\Rightarrow$ For each $y \in Y$, there exists a unique element $x \in X$ defined by $x = f^{-1}(y)$.

Example:

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x+1$.

This function is both 1-1 and onto.

$\Rightarrow$ It is a bijection and hence we can find f^{-1} .

Solution

Since $f(x) = 2x+1 \Rightarrow$

$$y = 2x+1$$

$$y-1 = 2x$$

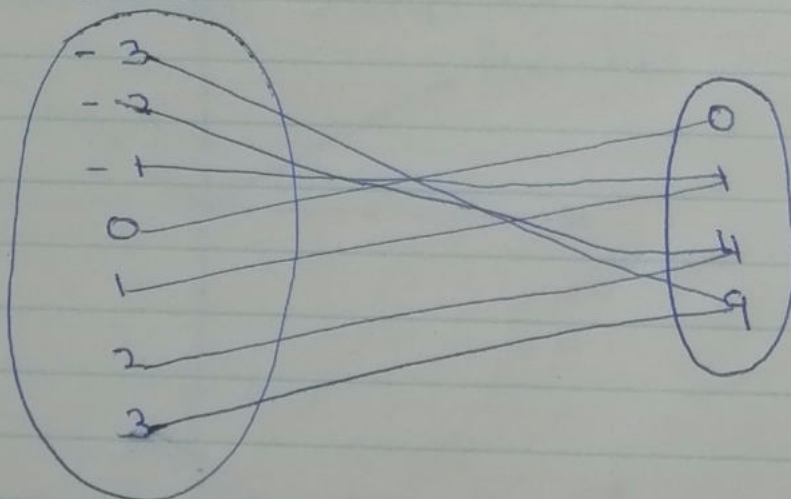
$$x = \frac{y-1}{2} \quad \text{[The inverse function]}$$

Consider the example.

Given $A = \{-3, -2, -1, 0, 1, 2, 3\}$

$B = \{0, 1, 4, 9\}$.

A function f is defined by $f: A \rightarrow B$ given by $f(x) = x^2$.

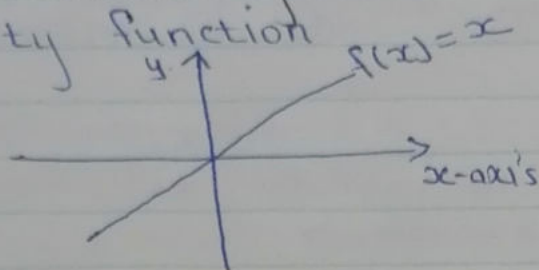


$\Rightarrow f$ is onto function.

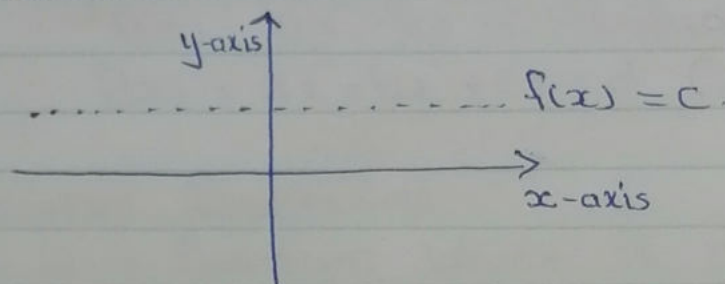
Some Basic Algebraic Functions

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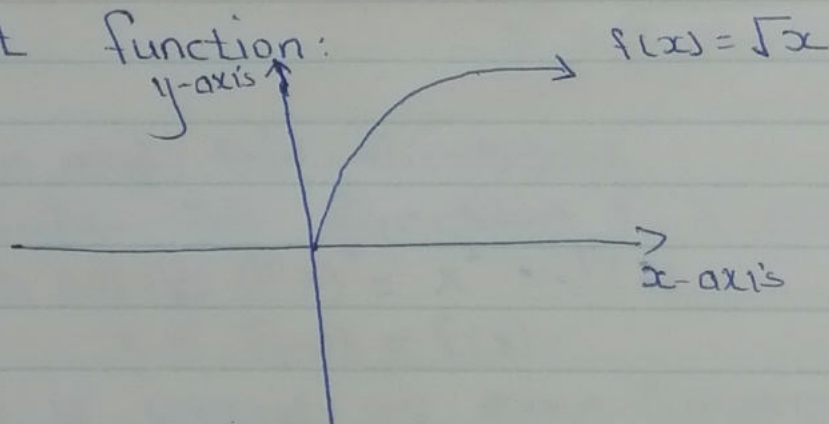
(a) Identity Function



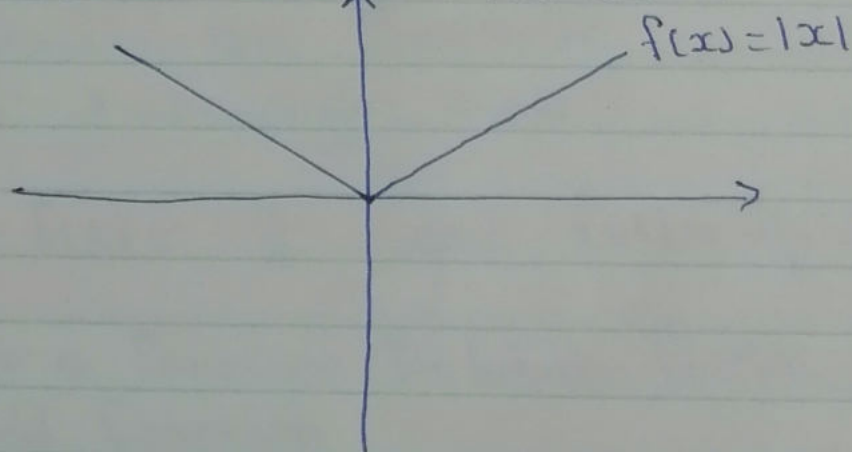
(b) Constant function



(c) Root function:



(d) Absolute value function



The range of $f(x) = |x|$ is $\mathbb{R}^+$

(e) Polynomial Functions

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Any function of the form

$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ where n is a positive integer or zero and a_i 's are fixed real numbers is called a polynomial function

e.g. $f(x) = a_1x + a_0$ (linear function)

$f(x) = ax^2 + bx + c \Rightarrow$ (Quadratic function)

$f(x) = ax^3 + bx^2 + cx + d$ (Cubic function)

(f) Rational Functions

This is a function which is a quotient of two functions that are polynomial

e.g. $f(x) = \frac{3x^3 + 2x + 1}{x^2 + x + 1}$

(g) Even and Odd Functions

Consider the function $f: x \longrightarrow x^2$

$\Rightarrow$ for $(-x) \Rightarrow f(-x)^2 = x^2 = f(x)$

i.e. $f(-x) = f(x)$

This is called an even function.

Even functions are not 1-1.

Consider the function

$f: x \longrightarrow \frac{1}{x}; x \neq 0$

$\Rightarrow f(x) = \frac{1}{x}$ and $f(-x) = \frac{1}{-x} = -1\left(\frac{1}{x}\right) = -f(x)$

For a function for which $f(-x) = -f(x)$ is an odd function.

(h) Composite Functions

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If f is a function from X to Y and g is a function from Y to Z , then the composite function $g \circ f$ is the function from X to Z defined by $(g \circ f)(x) = (g f(x))$ for every $x \in X$.

Similarly we have

$$(f \circ g)(x) = f(g(x)) \text{ for every } x \in X.$$

Example

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 1$ and $g(x) = 2x - 1$.

find

- (i) $f \circ g$ (ii) $g \circ f$ (iii) $g \circ f(3)$ (iv) $f \circ g(-2)$
(v) $f(3)$ (vi) $g(7)$.

Solution

(i) $f \circ g = f(g(x)) \Rightarrow$ The value of f at $g(x)$.

$$\begin{aligned} \text{Since } f(x) &= x^2 + 1 \Rightarrow f \circ g(x) = (2x - 1)^2 + 1 \\ &= 4x^2 - 4x + 1 + 1 \\ &= 4x^2 - 4x + 2. \end{aligned}$$

(ii) $g \circ f = g(f(x)) \Rightarrow$ The value of g at $f(x)$.

$$\begin{aligned} \text{Since } g(x) &= (2x - 1) \Rightarrow g(f(x)) = 2(x^2 + 1) - 1 \\ &= 2x^2 + 2 - 1 \\ &= 2x^2 + 1. \end{aligned}$$

(iii) $g \circ f(x) = g(f(x))$ at $x = 3$.

$$\Rightarrow g \circ f(3) = 2(3^2) + 1 = 19.$$

(iv) $f \circ g(x) \Rightarrow$ value of $f(g(x))$ at $x = -2$
 i.e. $4x^2 - 4x + 2$ at $x = -2$
 $= 4(-2)^2 - 4(-2) + 2 = 26.$

(v) $f(3) \Rightarrow$ The value of f at $x = 3$.
 i.e. $3^2 + 1 = 10$

(vi) $g(7) \Rightarrow$ The value of g at $x = 7$
 i.e. $2(7) - 1 = 13.$

Exercise:

1. If $f: x \rightarrow 2x+1$ and $g: x \rightarrow x-3$.

Determine the following functions

(i) $f \circ g$. Ans = $2x - 5$

(ii) $g \circ f$ Ans = $2x - 2$

(iii) $(f \circ g)^{-1}$ Ans $\frac{y+5}{2}$

(iv) $(g \circ f)^{-1}$ Ans $\frac{y+2}{2}$

(v) $f^{-1} \circ g^{-1}$ Ans $\frac{y+2}{2}.$

2. Find the domain and the range of the functions

(i) $f(x) = -4x + 3.$

(iv) $f(x) = \frac{1}{(1-4)^2}$

(ii) $f(x) = \sqrt{4-x^2}$

(v) $f(x) = |x-4|$

(iii) $f(x) = \frac{1}{x-4}$

(vi) $f(x) = \begin{cases} -5 & \text{if } x < -5 \\ x & \text{if } -5 \leq x \leq 5 \\ 5 & \text{if } x > 5 \end{cases}$